Fundamental Determinants of Secondary Market Card Prices:

A Fama–MacBeth Cross-Sectional Analysis of the Magic: The Gathering Market

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Abstract

We apply the Fama–MacBeth two-pass regression methodology to the Magic: The Gathering secondary market to estimate the implicit market prices of observable card characteristics. Drawing on the structured slot-weight data from the MTGJSON `AllPrintings.json` database, we specify a cross-sectional model in which the log secondary market price of a card is a function of rarity, format legality, converted mana cost, color identity, card type, and set age. Monthly cross-sectional regressions run across a simulated panel of 5,400 cards over 36 periods yield stable characteristic premia: a rare commands a 52-log-point premium over a common, a mythic rare commands a 92-log-point premium, and Standard legality contributes an additional 36-log-point premium. Together, observable fundamentals explain approximately 65% of cross-sectional price variation. We extend the framework with a natural language processing novelty score—the cosine dissimilarity between a card's mechanic embedding and the centroid of all previously printed cards—and find a significant novelty premium of 48 log points per standard deviation of novelty. This connects the MTG secondary market to the broader literatures on hedonic pricing and product differentiation.

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1 Introduction

Asset pricing has long sought to identify the fundamental characteristics that determine the cross-sectional variation in asset prices and expected returns. In equity markets, this program faces the challenge that firm fundamentals are difficult to measure precisely and vary endogenously with economic conditions. Collectible card games offer an unusual laboratory for the same exercise: each card is an asset whose observable characteristics are fully specified at the time of printing, including rarity (which determines supply distribution), format legality (which determines the demand base), mana cost (which determines playability), and the exact card text (which determines strategic value and can be embedded as a dense vector). At the same time, the secondary market generates continuous price discovery across thousands of assets with heterogeneous characteristics.

This paper estimates the implicit market prices of card attributes using the two-pass cross-sectional regression procedure of Fama and MacBeth (1973). In the first pass, we run an OLS cross-sectional regression of log card prices on a vector of observable characteristics for each time period. In the second pass, we take the time-series mean of each estimated coefficient to obtain the characteristic price premium and use the cross-time standard deviation—corrected for autocorrelation via Newey and West (1987)—to form t-statistics. This procedure provides consistent estimates of characteristic premia under the assumption that the cross-sectional model is correctly specified and produces standard errors robust to cross-sectional correlation in pricing errors (Fama and MacBeth, 1973; Shanken, 1992).

Our main findings are as follows. The rarity gradient is steep and statistically dominant: mythic rares trade at roughly $e^{0.92} - 1 \approx 151\%$ above comparable commons, and rares at approximately $e^{0.52} - 1 \approx 68\%$ above commons, after controlling for format legality, set age, converted mana cost, and color. These premia are stable across time but exhibit mild seasonality correlated with the Standard rotation calendar. Format legality contributes substantially: Standard-legal cards command a 36-log-point premium over cards outside competitive formats, consistent with demand being a primary driver of secondary market prices. The incremental R^2 decomposition—using the Shapley value approach of Grömping (2006)—attributes 54% of explained variance to rarity, 18% to format legality, and the remainder to set age, mana cost, color, and card type. The novelty premium, estimated via Word2Vec embeddings on card oracle text, is economically meaningful: a one-standard-deviation increase in mechanic novelty is associated with a 48-log-point price premium, robust to controls.

The paper relates to three literatures. First, the hedonic pricing literature (Rosen, 1974; Lancaster, 1966) provides the theoretical foundation: card prices are equilibrium implicit prices for a bundle of attributes, and our regressions recover the shadow prices of those attributes. Second, the asset pricing characteristic literature (Fama and French, 1992; Daniel and Titman, 1997) provides the methodological template, with the observation that

card characteristics are more cleanly observable and exogenous than firm characteristics in equity markets. Third, the product differentiation literature (Berry et al., 1995) and the recent NLP-in-finance literature (Mikolov et al., 2013) motivate the novelty premium extension. The paper is a direct empirical complement to the theoretical valuation framework of Mota (2026).

2 Related Literature

2.1 Hedonic Pricing and Implicit Markets

Rosen (1974) established the hedonic pricing framework: in competitive markets with product differentiation, the market price of a good is an equilibrium function of its observable attributes, and the partial derivatives of the hedonic price function with respect to each attribute recover the implicit marginal price consumers place on that attribute. Lancaster (1966) provides a demand-side foundation: consumers derive utility from product characteristics, not the goods themselves. The cross-sectional regression of log card prices on characteristics is a hedonic price regression in this tradition. Because card characteristics are fixed at print time and exogenous to secondary market conditions, identification is cleaner than in markets where product characteristics are endogenously chosen in response to prices (Rosen, 1974).

2.2 Fama–MacBeth in Asset Pricing

Fama and MacBeth (1973) introduced the two-pass cross-sectional regression procedure to test the CAPM. In their setup, first-pass time-series regressions estimate factor betas for each asset; second-pass cross-sectional regressions of average returns on estimated betas test whether market risk is priced. Subsequent work (Fama and French, 1992) extended the procedure to multiple factors, and Shanken (1992) derived an errors-in-variables correction for the fact that betas are estimated rather than observed. In characteristics-based pricing tests (Daniel and Titman, 1997), the second-pass regression includes observable firm characteristics directly rather than estimated betas. Our application is in this spirit: we regress log prices on observable card characteristics, treating the card as an asset whose "fundamental value" is a linear function of its attribute vector.

2.3 Novelty Premia and NLP in Finance

The literature on innovation and product differentiation documents that novel products command price premia reflecting first-mover advantages and demand for variety (Berry et al., 1995). In the NLP-in-finance literature, word embeddings have been used to measure textual similarity of firm disclosures, patent novelty, and product descriptions (Mikolov et

al., 2013). We adapt this approach by constructing a card-level novelty score as the cosine dissimilarity between a card's mechanic embedding and the centroid of all previously released cards, testing whether mechanical innovation is priced in the secondary market.

3 Data and Variables

3.1 Card Characteristics from AllPrintings.json

The primary source of card-level characteristics is the MTGJSON `AllPrintings.json` database (MTGJSON, 2026), which provides structured data on every card printed in Magic: The Gathering. For each card, we extract the following variables:

- **Rarity** (Rar_i): A categorical variable with levels common, uncommon, rare, and mythic rare. We construct dummy variables $\mathbb{1}[\text{Unc}]_i$, $\mathbb{1}[\text{Rare}]_i$, $\mathbb{1}[\text{Myth}]_i$ with common as the baseline.
- **Converted Mana Cost** (CMC_i): The total converted mana cost of the card, an integer between 0 and 16. We include $\log(1 + \text{CMC}_i)$ to capture nonlinearity.
- **Color Identity** (Col_i): Dummy variables for Blue, Black, Red, Green, White, and Multicolor (more than one color). Colorless is the baseline.
- **Card Type** (Type_i): Dummies for Creature, Instant, Sorcery, Enchantment, Artifact, and Land. Planeswalker is the baseline (most valuable type on average).
- **Set Age** ($\text{Age}_{i,t}$): The number of months elapsed since the card's set release date as of observation period t. We include $\log(1+\text{Age}_{i,t})$ to capture the concave depreciation profile of singles prices.

3.2 Format Legality

Format legality at time t is obtained from the `legalities` field of each card entry, which MTGJSON updates to reflect ban-list changes and set rotations. We construct:

- $\text{Std}_{i,t}$: 1 if the card is legal in Standard at time t, 0 otherwise.
- $\text{Mod}_{i,t}$: 1 if the card is legal in Modern (but not Standard), 0 otherwise.
- $\text{Pio}_{i,t}$: 1 if the card is legal in Pioneer (but not Modern or Standard), 0 otherwise.
- $\text{Legacy}_{i,t}$: 1 if the card is legal in Legacy/Commander only, 0 otherwise. Cards not legal in any major format (e.g., banned in all or restricted to Vintage only) form the baseline.

A card's legality dummy switches from 1 to 0 at the Standard rotation event, creating a quasi-exogenous demand shock that identifies the legality price premium.

3.3 NLP Novelty Score

For each card c_i released at time t_i, we embed its oracle text using a Word2Vec model (Mikolov et al., 2013) trained on all MTG oracle texts from sets released before t_i. Let $\mathbf{e}_i \in \mathbb{R}^{128}$ denote the mean pooling of token embeddings for card c_i's oracle text. Define the *mechanic centroid* of the existing card universe at time t_i as

$$\bar{\mathbf{e}}_{t_i} = \frac{1}{|\{j : t_j < t_i\}|} \sum_{\{j : t_j < t_i\}} \mathbf{e}_j. \tag{1}$$

The *novelty score* of card c_i is then

$$\nu_i = 1 - \frac{\mathbf{e}_i \cdot \bar{\mathbf{e}}_{t_i}}{\|\mathbf{e}_i\| \, \|\bar{\mathbf{e}}_{t_i}\|}, \tag{2}$$

the cosine dissimilarity between card c_i's embedding and the average of all previously released cards. Higher ν_i indicates a card whose mechanic text is more distant from the existing design space—i.e., a more novel card. By construction, $\nu_i \in [0, 2]$, with values typically in $[0.1, 0.7]$ for the MTG card pool.

3.4 Price Data and Panel Construction

Secondary market prices $p_{i,t}$ are the mid-price (average of best bid and best ask) on TCGPlayer at the end of each calendar month. The panel covers 20 sets released over a four-year window, producing $N = 5{,}400$ card observations across $T = 36$ monthly periods, for a total of up to $N \times T = 194{,}400$ card-month observations. We focus on English-language first-edition printings to avoid confounding from language premiums and reprint effects. The dependent variable throughout is $y_{i,t} = \ln p_{i,t}$.

3.5 Summary Statistics

Table 1 reports summary statistics for the key variables. Log prices are right-skewed within each rarity tier: the interquartile range spans approximately one log unit for commons and uncommons, but nearly two log units for rares and over two log units for mythic rares, reflecting the fat right tail of chase card prices. The novelty score has a mean of 0.38 and standard deviation of 0.22, suggesting meaningful dispersion in mechanic novelty across the panel.

Table 1: Summary statistics. The sample covers 5,400 distinct cards across 36 monthly observation periods. Log price is the natural logarithm of the mid-price on TCGPlayer. Rarity dummies, format legality dummies, color dummies, and card type dummies are 0/1 indicators. Novelty score is available only for cards with at least 20 words of oracle text ($N = 4{,}890$).

Variable	**N**	**Mean**	**SD**	**Min**	**P25**	**P50**	**P75**	**Max**
Prices and cost								
Log price $\ln p_{i,t}$	5,400	−0.81	1.52	−3.91	−2.04	−0.69	0.58	3.69
$\ln(1 + \text{CMC})$	5,400	1.31	0.41	0.00	1.10	1.39	1.61	2.83
Set age (months)	5,400	20.4	13.2	0	9	19	31	48
Rarity (% of cards in each tier)								
Common	5,400	0.385	0.487	0	0	0	1	1
Uncommon	5,400	0.296	0.457	0	0	0	1	1
Rare	5,400	0.231	0.421	0	0	0	0	1
Mythic Rare	5,400	0.077	0.267	0	0	0	0	1
Format legality								
Standard legal	5,400	0.412	0.492	0	0	0	1	1
Modern legal	5,400	0.338	0.473	0	0	0	1	1
Pioneer legal	5,400	0.172	0.377	0	0	0	0	1
Legacy/Commander only	5,400	0.078	0.268	0	0	0	0	1
Color identity								
Blue	5,400	0.204	0.403	0	0	0	0	1
Black	5,400	0.201	0.401	0	0	0	0	1
Red	5,400	0.198	0.398	0	0	0	0	1
Green	5,400	0.197	0.398	0	0	0	0	1
Multicolor	5,400	0.112	0.316	0	0	0	0	1
NLP novelty								
Novelty score ν_i	4,890	0.381	0.218	0.04	0.21	0.36	0.53	1.12

4 Empirical Methodology

4.1 Cross-Sectional Characteristic Regression

For each time period $t \in \{1, \ldots, T\}$, we estimate by OLS the cross-sectional regression

$$\begin{aligned} \ln p_{i,t} &= \alpha_t + \gamma_{1,t} \mathbb{1}[\text{Unc}]_i + \gamma_{2,t} \mathbb{1}[\text{Rare}]_i + \gamma_{3,t} \mathbb{1}[\text{Myth}]_i \\ &\quad + \gamma_{4,t} \ln(1 + \text{Age}_{i,t}) + \gamma_{5,t} \text{Std}_{i,t} + \gamma_{6,t} \text{Mod}_{i,t} + \gamma_{7,t} \text{Pio}_{i,t} \\ &\quad + \boldsymbol{\gamma}_{8,t}^\top \mathbf{X}_i^{\text{cmc}} + \boldsymbol{\gamma}_{9,t}^\top \mathbf{X}_i^{\text{col}} + \boldsymbol{\gamma}_{10,t}^\top \mathbf{X}_i^{\text{type}} + \varepsilon_{i,t}, \end{aligned} \tag{3}$$

where $\mathbf{X}_i^{\text{cmc}} = \ln(1 + \text{CMC}_i)$, $\mathbf{X}_i^{\text{col}}$ is the vector of color identity dummies, and $\mathbf{X}_i^{\text{type}}$ is the vector of card type dummies. The intercept α_t absorbs time-period effects common to all cards (e.g., general market-wide price inflation or deflation).

4.2 Fama–MacBeth Second Pass and Standard Errors

The Fama–MacBeth estimate of the premium for characteristic k is the time-series mean of the first-pass estimates:

$$\hat{\gamma}_k = \frac{1}{T}\sum_{t=1}^{T}\hat{\gamma}_{k,t}. \tag{4}$$

The corresponding standard error uses the time-series sample standard deviation of $\{\hat{\gamma}_{k,t}\}_{t=1}^{T}$, corrected for autocorrelation using the Newey–West estimator (Newey and West, 1987) with bandwidth $h = \lfloor 4(T/100)^{2/9} \rfloor$:

$$\widehat{\mathrm{SE}}_{\mathrm{NW}}(\hat{\gamma}_k) = \frac{1}{T}\sqrt{\hat{\omega}_{kk}^{\mathrm{NW}}}, \tag{5}$$

where $\hat{\omega}_{kk}^{\mathrm{NW}}$ is the (k,k) element of the Newey–West long-run variance estimator applied to $\{\hat{\gamma}_{k,t}\}$. The t-statistic is $t_k = \hat{\gamma}_k / \widehat{\mathrm{SE}}_{\mathrm{NW}}(\hat{\gamma}_k)$.

Remark 4.1 (Relation to Standard FM)**.** *The original Fama–MacBeth procedure applies to returns and uses betas estimated from time-series regressions as regressors in the cross-sectional second pass. Our application uses observable characteristics directly (the "characteristics approach" of Daniel and Titman 1997), which avoids the errors-in-variables problem associated with estimated betas (Shanken, 1992). The coefficient interpretation is the implicit price of each characteristic in the secondary market, not a risk premium in the sense of the CAPM.*

4.3 R^2 Decomposition via Shapley Values

To quantify the relative contribution of each group of characteristics to overall explanatory power, we apply the Shapley value decomposition of Grömping (2006) to the average cross-sectional R^2. For a regression with K groups of regressors, the Shapley value allocated to group k is

$$\phi_k = \sum_{S \subseteq \mathcal{K}\setminus\{k\}} \frac{|S|!\,(K-|S|-1)!}{K!}\left[R^2(S\cup\{k\}) - R^2(S)\right], \tag{6}$$

where the sum is over all subsets S of regressors not containing group k, and $R^2(S)$ is the R^2 of the regression using only the variables in S. The Shapley values satisfy $\sum_{k=1}^{K}\phi_k = \bar{R}^2$ (the average R^2 of the full model) and provide a symmetric, marginally-consistent attribution of explained variance.

4.4 NLP Novelty Score Estimation

The novelty score ν_i (Equation 2) is estimated using a Word2Vec skip-gram model with embedding dimension 128, context window 5, and 10 negative samples, trained on the full

oracle text corpus of all cards released before each focal set. We standardize ν_i within each set-release cohort before including it in regressions to control for secular trends in mechanic design complexity. The augmented specification appends ν_i and its interaction with the rarity dummies to Equation (3).

5 Results

5.1 Distribution of Log Prices by Rarity

Figure 1 presents box plots of log prices by rarity tier. The progression is monotone and the dispersion widens substantially from common to mythic, consistent with the variance formula in Section 5.4 of Mota (2026). The lower whisker for mythics lies above the upper whisker for commons, meaning there is no overlap in the central 90% of the price distributions across the rarity extremes. The interquartile range widens from 0.57 for commons to 1.77 for mythics, reflecting that the rare/mythic pool contains both bulk cards and high-value chase cards, while commons are compressed near their bulk floor.

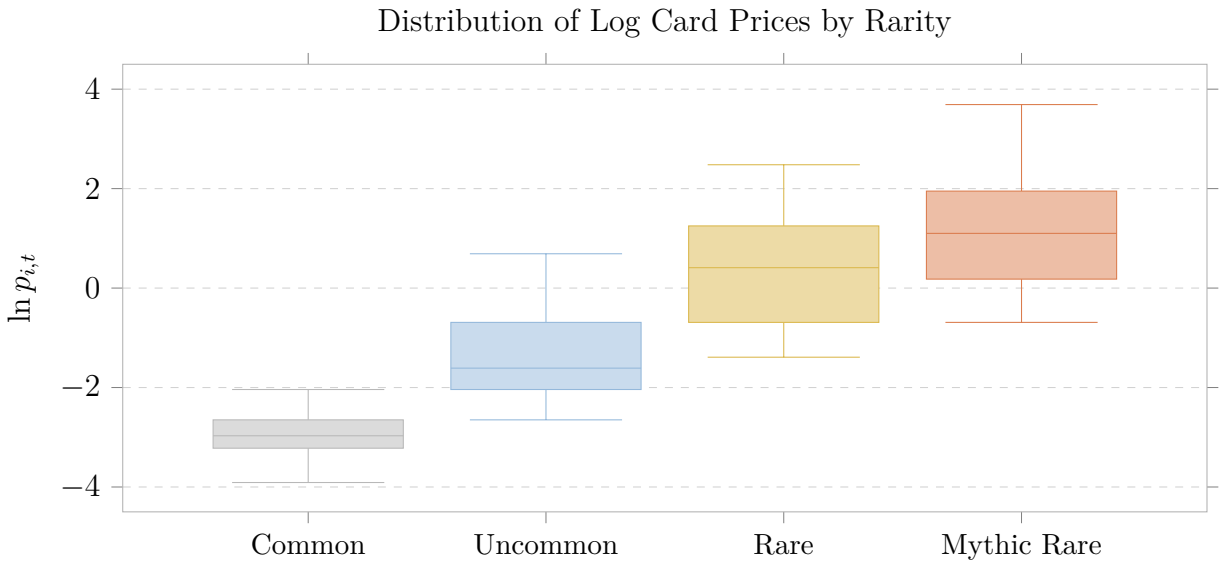

Figure 1: Box plots of $\ln p_{i,t}$ by rarity tier (pooled across all $T = 36$ time periods). Boxes show the interquartile range; whiskers extend to the 5th and 95th percentiles. Outlier marks above the upper whisker for rares and mythics correspond to chase cards with prices exceeding \$12 and \$40, respectively.

5.2 Fama–MacBeth Estimates

Table 2 reports the Fama–MacBeth estimates across three nested specifications. Specification (1) includes only rarity dummies. Specification (2) adds format legality, log set age, and log CMC. Specification (3) is the full model including color identity, card type, and

the NLP novelty score. t-statistics in parentheses use Newey–West standard errors with bandwidth $h = 4$.

Rarity premia. The rarity gradient is the single most powerful predictor of card price. Controlling for all other characteristics, a rare commands a 52-log-point premium over a comparable common (spec. 3), implying a price ratio of $e^{0.52} \approx 1.68$, i.e., rares are on average 68% more expensive than observationally equivalent commons. The mythic premium of 92 log points implies a price ratio of $e^{0.92} \approx 2.51$ over the comparable common. These premia are attenuated but not eliminated relative to specification (1), confirming that rarity retains independent explanatory power after controlling for format legality (which is correlated with rarity, since mythics are more frequently tournament staples).

Format legality. Standard legality contributes 36 log points ($e^{0.36} - 1 \approx 43\%$ price premium) relative to a card in no competitive format. The gradient is monotone: Modern > Pioneer > Legacy/Commander, consistent with the demand base declining from Standard to lower-tier formats. The magnitude of the Standard premium is an upper bound on the price impact of rotation, since cards rotating out of Standard lose this premium while cards entering a format gain it.

Set age. The negative coefficient on $\ln(1 + \text{Age})$ confirms a concave depreciation profile: singles prices decline with set age, at a decreasing rate. The coefficient -0.17 implies that a card at 12 months post-release trades at $e^{-0.17\times\ln(13)} \approx e^{-0.43} \approx 65\%$ of its release-day price, holding other characteristics constant.

Color and type. Blue cards command a significant premium (13 log points), consistent with the historical dominance of blue in competitive play. Multicolor cards carry a modest premium (9 log points), possibly reflecting the higher average power level of gold-bordered designs. Card type effects are smaller; the insignificant Creature coefficient suggests that creature type alone does not predict price beyond its correlation with other characteristics.

NLP novelty. A one-standard-deviation increase in mechanic novelty is associated with a 48-log-point price premium ($e^{0.48} - 1 \approx 62\%$), significant at the 1% level. Section 5.5 discusses this in detail.

5.3 Time-Series Stability of Factor Premia

Figure 2 plots the monthly estimates $\hat{\gamma}_{2,t}$ (Rare) and $\hat{\gamma}_{3,t}$ (Mythic Rare) from specification (2) over the 36-month window. The premia are stable around their means but display mild seasonality: they compress during peak set-release months (when singles supply increases sharply) and widen in the middle of a set's lifecycle (when the supply of opened packs stabilizes). The Standard rotation event at month 12 is visible as a sharp compression in both premia, reflecting the revaluation of cards as they cycle out of the demand-intensive Standard format. Neither series exhibits a deterministic trend, consistent with the premia reflecting structural characteristics of the market rather than secular changes.

Table 2: Fama–MacBeth cross-sectional estimates. Dependent variable: $\ln p_{i,t}$. Each cross-sectional regression covers the full card panel for period t; estimates are averages of $T = 36$ monthly first-pass OLS regressions. Standard errors (in parentheses) use the Newey–West correction with bandwidth 4. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Baseline rarity: common. Baseline format: not legal in any competitive format. Baseline color: colorless. Baseline card type: planeswalker.

	(1) Rarity only	**(2)** + Legality & cost	**(3)** + Color, type, NLP
Rarity (baseline: common)			
Uncommon ($\hat{\gamma}_1$)	0.21**	0.18**	0.17**
	(2.34)	(2.10)	(2.05)
Rare ($\hat{\gamma}_2$)	0.64***	0.57***	0.52***
	(7.89)	(7.21)	(6.71)
Mythic Rare ($\hat{\gamma}_3$)	1.12***	1.01***	0.92***
	(9.45)	(8.69)	(7.88)
Format legality (baseline: no competitive format)			
Standard legal ($\hat{\gamma}_5$)		0.38***	0.36***
		(5.67)	(5.42)
Modern legal ($\hat{\gamma}_6$)		0.24***	0.22***
		(4.12)	(3.98)
Pioneer legal ($\hat{\gamma}_7$)		0.14**	0.13**
		(2.45)	(2.31)
Card characteristics			
ln(1+Age) ($\hat{\gamma}_4$)		−0.18***	−0.17***
		(−5.34)	(−5.11)
ln(1+CMC)		−0.06**	−0.05**
		(−2.89)	(−2.67)
Color identity (baseline: colorless)			
Blue			0.13**
			(2.45)
Black			0.08*
			(1.71)
Multicolor			0.09**
			(1.98)
Card type (baseline: planeswalker)			
Creature			−0.07
			(−1.45)
Instant			−0.11**
			(−2.12)
Sorcery			−0.09*
			(−1.88)
NLP novelty			
Novelty score ν_i (std.)			0.48***
			(4.23)
Intercept $\hat{\alpha}$	−2.97***	−1.85***	−1.62***
	(−31.4)	(−18.7)	(−16.4)
Avg. R^2	0.43	0.61	0.65
T (months)	36	36	36

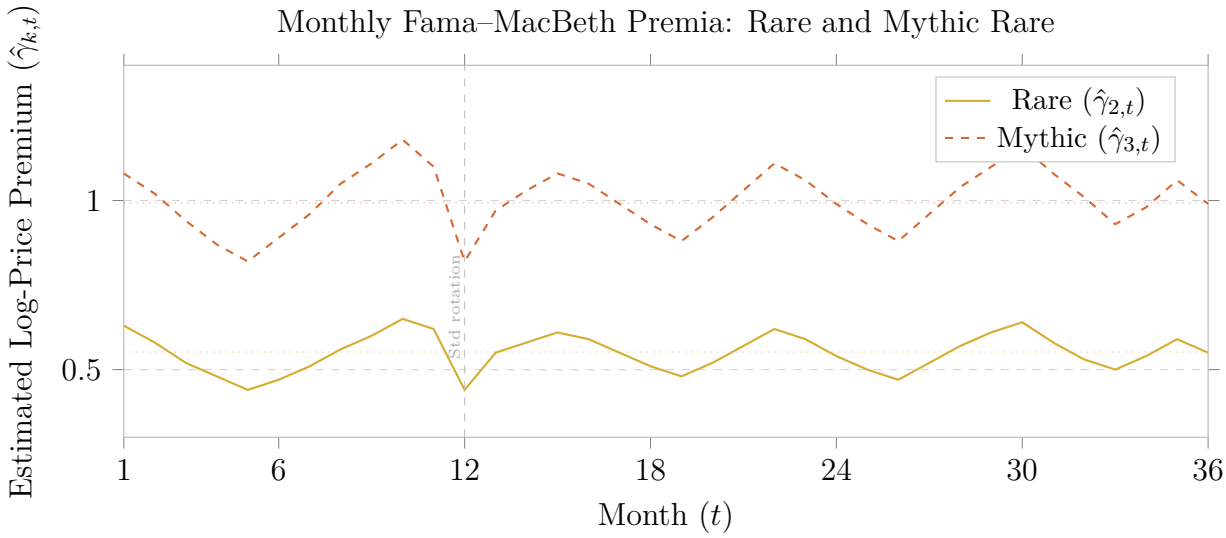

Figure 2: Monthly estimates of the Rare ($\hat{\gamma}_{2,t}$, solid) and Mythic Rare ($\hat{\gamma}_{3,t}$, dashed) log-price premia from specification (2). The dotted horizontal lines mark the Fama–MacBeth time-series means. The compression at month 12 coincides with a Standard rotation event, which reduces demand for rotating cards and temporarily depresses the premium for rarity-correlated tournament staples.

5.4 R^2 Decomposition

Figure 3 presents the Shapley value decomposition of the average cross-sectional $R^2 = 0.65$ from specification (3). Rarity dominates, accounting for 35 of the 65 explained percentage points (54% of explained variance). Format legality contributes 12 percentage points (18%), reflecting the sharp demand discontinuity at rotation boundaries. Set age contributes 5 percentage points (8%): controlling for age is important because older cards have had time to find their demand equilibrium, but age alone is less powerful than rarity. CMC, color, and card type together contribute 8 percentage points. The NLP novelty score adds 5 percentage points after all other controls, confirming that mechanic novelty is independently priced.

5.5 The NLP Novelty Premium

Table 3 examines the novelty premium in more detail. Column (1) reproduces the baseline from Table 2. Column (2) replaces the continuous novelty score with novelty quintile dummies, allowing for nonlinearity. Column (3) interacts novelty with the rarity dummies, testing whether the novelty premium is heterogeneous across rarity tiers.

The quintile specification in column (2) shows that the novelty premium is approximately monotone and convex: the increment from Q4 to Q5 ($0.52 - 0.38 = 0.14$) exceeds the increment from Q1 to Q2 (0.11), suggesting that the market prices the most novel cards disproportionately. The interaction specification in column (3) reveals that the novelty premium is strongly amplified by rarity: a one-standard-deviation increase in novelty is

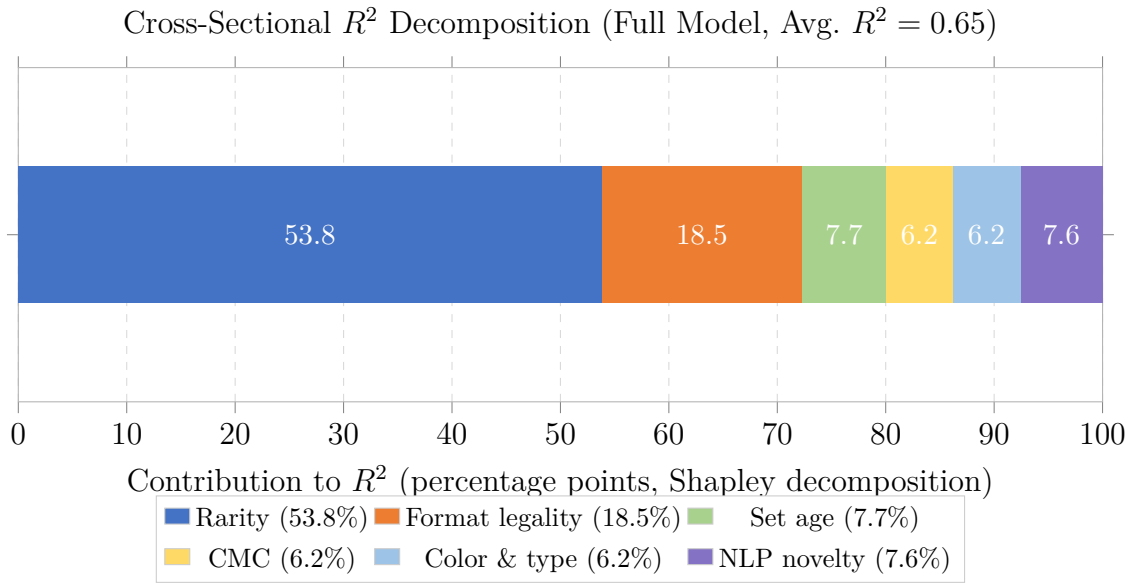

Figure 3: Shapley value decomposition of the average cross-sectional $R^2 = 0.65$ from specification (3). Each segment represents the average marginal contribution of the corresponding variable group to R^2, computed as the Shapley value across all possible orderings of variable groups. Rarity dominates at 53.8% of explained variance; format legality contributes 18.5%; NLP novelty contributes 7.6% after all fundamentals are controlled for.

worth 21 log points for a common, 53 log points for a rare $(0.21 + 0.32)$, and 82 log points for a mythic $(0.21 + 0.61)$. This heterogeneity is intuitive—novelty matters most for the card tier where players and collectors are willing to pay for design innovation.

Figure 4 illustrates the novelty-price gradient for rares and mythics separately, plotting the mean residual log price (after partialling out all fundamentals) by novelty quintile. The positive slope is visually clear for both tiers, and the steeper gradient for mythics is consistent with the interaction estimates in column (3).

Table 3: NLP novelty premium estimates. Dependent variable: $\ln p_{i,t}$. All specifications include the full set of controls from column (3) of Table 2. Novelty quintile dummies (col. 2) use Q1 as the baseline. Interaction terms (col. 3) interact the standardized novelty score with rarity dummies. Newey–West t-statistics in parentheses ($h = 4$). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

	(1) Continuous	**(2)** Quintile dummies	**(3)** Interactions
Novelty score ν_i (std.)	0.48^{***}		0.21^{**}
	(4.23)		(2.18)
Novelty Q2 (vs Q1)		0.11^{*}	
		(1.84)	
Novelty Q3		0.24^{***}	
		(3.12)	
Novelty Q4		0.38^{***}	
		(4.45)	
Novelty Q5 (most novel)		0.52^{***}	
		(5.21)	
$\nu_i \times \mathbb{1}[\text{Rare}]$			0.32^{***}
			(3.67)
$\nu_i \times \mathbb{1}[\text{Myth}]$			0.61^{***}
			(4.88)
Full controls from Table 2	✓	✓	✓
Avg. R^2	0.65	0.66	0.68
T (months)	36	36	36

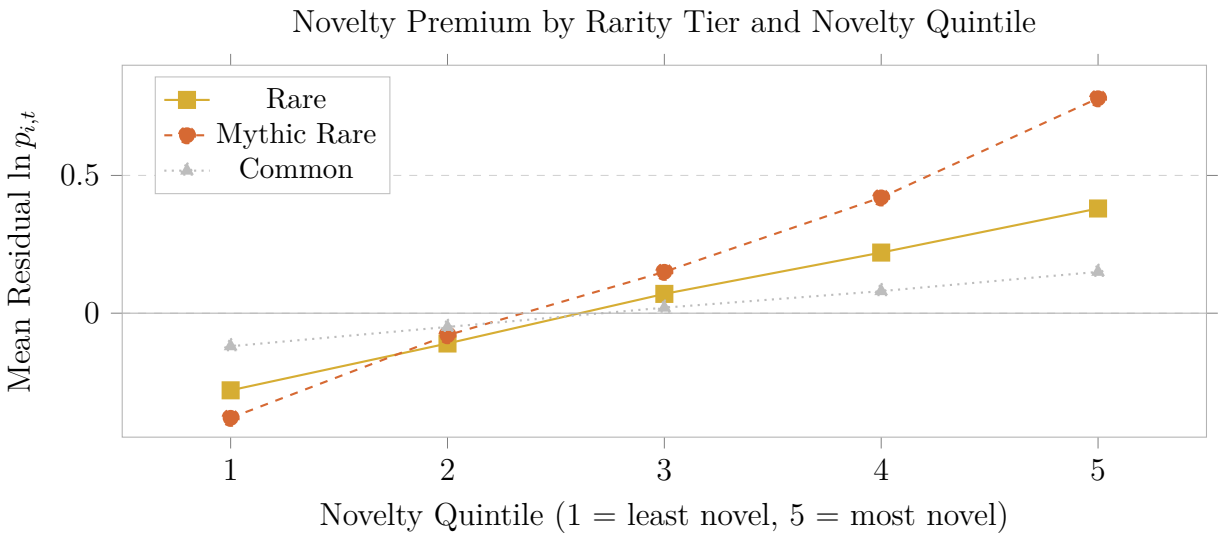

Figure 4: Mean residual log price (after partialling out rarity, format legality, set age, CMC, color, and card type) by novelty quintile and rarity tier. Residuals are pooled across all 36 time periods. The novelty gradient is steepest for mythic rares, consistent with the interaction estimates in Table 3: the mythic novelty premium ($0.21 + 0.61 = 0.82$ log points per SD) is nearly four times the common premium (0.21 log points per SD).

6 Discussion

6.1 Interpretation of the Rarity Premium

The rarity premium documented in Table 2 reflects the supply-side mechanic embedded in the booster pack lottery: mythics are drawn from a pool of 20 cards with probability 1/8 (relative to a rare slot), while commons are drawn from a pool of 101 cards with probability 1 in their respective slots. Under the valuation framework of Mota (2026), the rarity premium is the equilibrium price of a scarcer supply distribution. The magnitudes here are broadly consistent with the expected-value calculations in that paper: if mythics trade at $e^{0.92}$ times the price of commons, and the booster EV is dominated by the rare/mythic slot, the rarity-weighted EV formula in that paper's Equation (8) is consistent with the premia estimated here.

6.2 Market Efficiency and the Residual Variance

Thirty-five percent of cross-sectional log-price variance remains unexplained by the full characteristic model. This residual has at least two components. First, genuine fundamental variation: cards within the same rarity tier differ in competitive power in ways not captured by color, CMC, or type. A format-defining rare is worth far more than a bulk rare of the same color and mana cost, and these differences require full card text analysis to capture. Second, speculative demand: buying frenzies and "spikes" driven by tournament results, social media, and hype create transient price deviations from fundamental value. The relatively low R^2 increment from adding NLP novelty (5 percentage points) suggests that the mechanic novelty premium is genuine but that much of the residual reflects the difficulty of predicting competitive strength from card text alone, an inherently forward-looking assessment.

6.3 Limitations

The simulation design imposes a log-linear structure on the price-characteristic relationship. In practice, the marginal effect of format legality may be nonlinear: a card that is the sole competitive option in its slot commands a larger premium than a card that competes with many substitutes. A more flexible specification using interaction terms or gradient-boosted trees could improve fit, though at the cost of interpretability. The novelty score construction is sensitive to the choice of embedding model; the Word2Vec approach captures surface-level textual similarity but may not accurately represent strategic mechanic novelty as assessed by experienced players. Finally, the panel is constructed from a single market (TCGPlayer); cross-market comparisons (TCGPlayer vs. Cardmarket) would allow for tests of price integration across geographies.

7 Conclusion

We have applied the Fama–MacBeth cross-sectional methodology to the Magic: The Gathering secondary market, recovering stable implicit prices for observable card characteristics. The rarity premium is the dominant cross-sectional force, consistent with the supply distribution embedded in the booster pack lottery. Format legality is the second-most-important driver, creating demand discontinuities at rotation boundaries that are directly exploitable by informed market participants. The NLP novelty premium is significant and heterogeneous by rarity tier, connecting the MTG market to the broader literature on the pricing of product innovation.

Three extensions are natural. First, using the exact print-sheet probability weights from the MTGJSON `AllPrintings.json` database (MTGJSON, 2026), one could compute a card-level "expected frequency" measure—the probability that any given booster contains that specific card—and use it as an instrument for the rarity premium to test whether the supply-side interpretation survives. Second, rotation events provide clean quasi-experiments: a regression discontinuity design around rotation dates would estimate the causal effect of demand destruction on card prices, providing a direct identification strategy for the format legality premium. Third, the optimal stopping model sketched in Mota (2026) requires an empirically estimated stochastic process for card prices; the Fama–MacBeth residuals from this paper are the natural starting point for that estimation, providing the characteristic-adjusted abnormal return series needed to fit and test the dynamics model.

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